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Digital Twins Technology for Smart City Development. A Case Study of Poland's Largest Cities

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Abstract

Smart city development aims to build urban areas providing a high quality of life through modern infrastructure. Local governments have different strategic priorities. In particular, they focus on socio-economic capabilities, technological benefits, and sustainability goals. These areas are interrelated through spatial, time, and infrastructural dependencies. Intelligent city management could benefit from the simulation and visualization of these relationships. A digital twin is a solution that can oversee the complexity of city infrastructure spanning multiple locations, IoT devices, and data from other systems. The essence of digital twins is a bi-directional mapping of the relationship between physical and virtual space. The article aims to identify the state of knowledge about digital twins among local governments and the areas and possibilities using this technology to support city management. To achieve these goals, the authors conducted a research procedure covering the following stages: (1) identifying the directions of application of digital twins in smart cities based on the literature; (2) Conducting interviews with local governments representatives of the ten largest cities in Poland; (3) Expert assessment of potential application areas of digital twins, taking into account the needs, existing infrastructure, and information resources. The paper's contribution is recognizing the areas in city management that can benefit from digital twin implementation.

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1. Introduction

The aim of a smart city is to use technology and data to improve the quality of life for citizens, enhance sustainability, and increase the efficiency of urban services and infrastructure. Various systems are implemented to support the development of smart city solutions. Local governments have different strategic priorities. In particular, they focus on socio-economic capabilities (e.g., economic opportunities, cultural richness, and demographic-geographical balance), technological benefits (e.g., healthcare, education, government), and sustainability goals (e.g., clean, green, safe, and secure environments) [1]. It should be noted that technological solutions are often related to the legal aspects of user safety, confidentiality, and personal data protection [2]. These areas are interrelated through spatial, time, and infrastructural dependencies. Therefore, integrated smart city management could benefit from the simulation and visualization of these relationships. The use of modern ICT solutions in smart city management is addressed by the authors on several occasions [3][4][5].

A digital twin is a solution that can oversee the complexity of city infrastructure networks spanning multiple locations, thousands of IoT devices, and data from on-premises, centralized systems that still exist in the cities' IT infrastructures. The essence of digital twins is a bi-directional mapping of the relationship between physical and virtual space [6]. A change in the state of the physical object leads to a change in its digital equivalent. Digital twins also enable virtual objects to control physical entities without human intervention [7]. Observing the market of IT systems, it can be noticed that software companies already offer solutions for cities, advertising them as Digital Twins. Their primary purpose is to combine large amounts of available data in a standard, easy-to-understand interactive platform. The 3D or 4D visual model helps government agencies and other stakeholders to make better decisions about their projects or assets. Creating a versatile framework requires an individual approach to integrating existing data sources and the systems operating in a given city.

Several critical challenges must be identified and addressed to ensure successful deployment and operational effectiveness in implementing digital twins in smart cities [8][9][10]. These challenges can be categorized into technical, security, financial, and organizational aspects, each presenting unique hurdles (Fig.1). It is worth noting that these challenges are deeply interrelated, impacting one another in complex ways that demand a holistic approach to management and resolution.

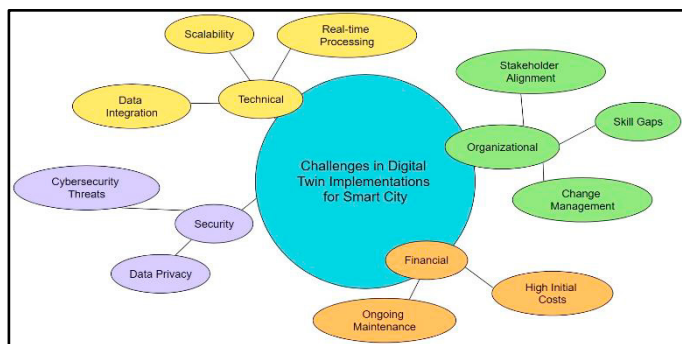


Fig 1. Challenges in Digital Twin implementations.

Among the mentioned categories, the financial aspect is least discussed within academic literature. The relative lack of discussion on the financial challenges of digital twin implementations in smart cities can be attributed to several factors. First, the complexity and variability of costs associated with these projects make it difficult to generalize financial models applicable across different contexts. Secondly, research often focuses more on advancing technological and theoretical aspects rather than examining practical financial implications. Additionally, detailed public data on expenditures are typically scarce due to commercial sensitivities, hindering thorough financial analysis [11]. Also, as the field of digital twins in smart cities is still emerging, research tends to prioritize foundational and operational issues over mature discussions of financial management. Lastly, there is a tendency to concentrate on

policy and strategic benefits, viewing financial considerations as secondary to the primary goal of technological integration.

The cognitive purpose of the article is to identify the state of knowledge about digital twins among local governments of the largest Polish cities and the areas and possibilities using this technology to support the management of a smart city.

The paper's theoretical contribution is recognizing the areas in city management that can benefit from digital twin implementation. The study's practical implications show the possibilities of semantic linkage between heterogeneous data sources, IoT networks, legacy management information systems, and web resources for supporting intelligent city management.

The paper is structured as follows: Section one examines how this work relates to existing research and what new insights it will contribute through the literature review. Section three presents the research methodology, followed by the results and discussion.

2. A literature overview

Solutions referred to as digital Twins have been undergoing an evolution in recent years. Since the term first appeared (2002), they were just ideas and theories. Today, the concept has become a practical solution in many business and industry sectors, particularly Industry 4.0. Before the advent of digital twin technology, smart city management relied on various technologies such as sensor networks, internet of things (IoT) devices, cloud computing, big data analytics, and artificial intelligence (AI). These technologies were used to collect and analyze data on various aspects of the city, such as traffic, energy consumption, air quality, and waste management. Urban planners used Computer-Aided Design (CAD) technology and smart maps powered by geospatial analytics. However, data conversion between CAD and GIS posed a significant challenge [12]. Integrating these technologies into a comprehensive digital twin system has enabled more accurate and efficient simulation, prediction, and optimization of city operations.

According to [13] we can categorize DT into three variants: digital model, digital shadow, and digital twin. In our opinion, the described solutions can be considered stages of technological maturity related to different types of data sharing between physical and digital systems. The digital model is a virtual representation of an existing physical system or its theoretical model, including its detailed description. The digital model can be used for the simulation of the physical system; it cannot have a direct influence on the functioning of the physical system. However, the simulation results are potentially useful for managing the physical system. The digital model can be the first step towards Digital Twin development as presented by [14]. The Digital Shadow augments the functionality of the digital model by offering a one-way data flow, enabling the real-time reflection of changes in the physical system in its virtual representation. The Digital Shadow is, therefore, useful for monitoring and early detection of emerging problems in the physical system. The Digital Shadow is a core component of Digital Twin [15]. A case study of Songdo (South Korea) presents the digital city shadow using data from surveillance cameras in public areas, interlinked traffic infrastructure, measuring instruments, and housing units [16]. It should be noted that the model functioning in Songdo is based on open data that residents have access to using functional applications. The city promotes the effective use of data as a marketing tool, trying to attract potential residents and businesses.

The digital twin encompasses the capabilities of the previously described models, adding two-way data linkage between the physical and virtual models. The advantage is the ability to affect the physical system by manipulating its digital representation to achieve the desired outcomes.

In our opinion, the three solutions described can be considered stages of technological maturity toward a fully functioning Digital Twin platform. They are related to different types of data sharing between physical and digital systems. Each stage requires the use of more and more advanced technologies.

From the literature review, it can be seen that there is no clear distinction between the digital shadow and digital twin concepts. Many literature review studies reveal that the concept and scope of DT lack a clear universal definition [17]. According to some authors, systems that visualize and allow simulations of urban space can be considered digital twins, and the simulation effects can be used in practice. The previously mentioned city of Songdo allows residents to control their homes using an app, so this solution can be classified as a Digital Twin. In turn, the case of the city of Zurich, described by Gerhard Schrotter and Christian Hürzeler [18] as a Digital Twin, does not indicate ways to control

the infrastructure automatically. The authors rather focus on the possibility of using an accurate digital representation of the city in planning its architecture. Zurich uses its virtual city model in the areas of environment (e.g., noise, air pollution and mobile phone radiation modeling), energy (e.g., solar potential analyses), urban planning (visualization of construction projects, shadow and visibility analyses), and in other areas. In this case, changes to the virtual model are not automatically reflected in real-world objects.

The importance of using digital twins can be measured in the added value to society [19], which is a primary goal of the Smart City concept. Therefore the data-driven 3D city modeling based on control systems, sensor networks, and urban artifacts only has attracted criticism because of its focus on the physical dimensions of cities [20]. Therefore, creating a holistic Digital City Twin requires a social focus, including data from social networks, demographic databases, and user feedback [21]. A comprehensive, scalable semantic approach should consider interdisciplinary knowledge of sociology, anthropology, and ethics studies [22].

The Digital City Twin (DCT) concept can be realized on three main levels [23]:

- Unit level
- System level
- System of systems level.

Unit level - means digital mapping of individual objects in the city (buildings, structures, infrastructure elements, etc.). The DCT in this case allows not only the digital mapping of the object and its records, but also enables simulation of its operation, prediction of behavior and analysis of results. These conclusions can be translated back into a physical object.

System level - is the combination of objects belonging to the same class or family into unified systems, such as the city's transportation system, land, and real estate, investments, and residents (social system).

System of systems-level (Process level) - is the merging of systems into one large digital urban organism; at this level, it is possible to manage the relationships between systems, pointing out interactions and interdependencies; this approach in a digital environment offers the possibility to support management processes.

Considering DT systems in the context of a smart city, the authors focus on various applications related to city management. The number of involved stakeholders increases in developing DT encompassing a wide range of domains. The Cambridge Digital Twin (CDT) project described by [23] aims to link various models and existing workflows in local governments to support their cooperation in shaping politics. Another Herrenberg (Germany) study focuses on supporting communication and decision-making between politicians, responsible administrative staff, experts from practice and different disciplines, and citizens [24]. The authors propose virtual and augmented reality applications to support collaborative and participatory urban planning, design, and decision-support processes.

An important aspect is a semantic link between different areas, which allows for analyzing their interactions. The critical problem, next to technological issues, is defining the responsibilities concerning decision-making and coordination of collaborative relationships between decision-makers. The efforts are also necessary to convince the stakeholders to provide and share their data on the DT platform and make appropriate legal arrangements [16].

3. Research procedure

This paper attempts to evaluate the state of knowledge about Digital Twins among local governments of the largest Polish cities and assess the potential of this technology in city management. To achieve these goals, the authors conducted a research procedure covering the following stages:

1. Determining the existing level of awareness and knowledge about digital twins among the local governments of the largest Polish cities. This includes understanding to what extent they are currently being implemented or considered.
2. Exploring the various applications of digital twin technology in smart city management. This involves reviewing the latest literature to categorize and describe how digital twins can enhance urban systems and services, focusing on sectors such as transportation, utilities, public safety, and environmental monitoring.

3. Assessing the recognition and understanding of potential ethical concerns associated with digital twins among local government officials. Key issues include data privacy, security, and ensuring equitable access to the benefits of this technology.
4. Implementing a series of guided interviews with representatives from the local governments of the ten largest cities in Poland. These interviews aimed to gather qualitative insights into the perceptions, expectations, and planned initiatives related to digital twins.
5. Engaging experts to evaluate the potential areas for digital twin application, considering the specific needs, existing infrastructure, and available information resources of Polish cities. This step aims at tailoring recommendations for digital twin deployments that are feasible and beneficial within the local contexts.

A survey was conducted on a selected group of localities to determine the potential of Digital Twins technology for city management. It was assumed that new technologies, usually cost-intensive, are first implemented in cities with high potential, derived from the number of residents, the city's role in the region, and financial capabilities. Due to the familiarity with the procedures of city offices and the lack of cultural and language differences, the survey was conducted among the largest cities in Poland. The largest city is the capital Warsaw, with a population of 1,860,000. It is the clear leader among Polish cities. The potential of no other city can be compared with Warsaw. The next city on the list is Krakow, with a population of about 800,000. Further down the list of the top 10 are cities with populations of just over 200,000.

For the study, it was initially decided to analyze the ten largest cities in Poland. Eventually, 16 cities were selected for the study. The list was additionally supplemented with smaller cities (less than 200,000 residents) that are provincial capitals or have offices that are important to the operation of the provinces. Such a small number of cases will not allow the application of statistical tools. The conclusions and results are more like an extended case study.

By existing procedures in the offices, the online survey questionnaire was sent to the offices of city secretaries in the form of a link (<https>). The secretary decides on the workflow process and indicates the organizational unit to complete the survey. Most offices responded to the correspondence, however, indicating that current regulations do not allow online forms to be filled out (most likely to avoid the risk of opening unverified links to websites). Therefore the online form was, in most cases, converted into .doc or .pdf document. Finally, 12 completed forms were received out of 16 sent out. The questionnaires sent in the form of .doc and .pdf documents contained an additional field for the respondents' comments that would not have been possible with the online surveys.

4. Research results

The first conclusion is not directly derived from the survey questions. It concerns the senders and organizational units of the cities that sent the completed documents. The study revealed that all large cities in Poland have special departments responsible for developing strategies and monitoring investments involving IT solutions. In 6 cases (out of 12 completed), the names of the organizational units include the terms: Smart City, IT, digitization, and digital projects. This first conclusion indicates that all large cities in Poland realize the importance of modern IT solutions in Smart City management. Offices monitor new technologies and implement them as far as organizational, financial, and technical possibilities allow.

The first part of the survey concerned familiarity with the term "digital twin." This concept, being relatively new, is not yet widely recognized. However, the responses received indicate that all respondents are familiar with it and correctly identify the features and capabilities of digital twins.

Respondents were further asked about the extent of the use of digital twins or their plan to use this technology. Most responses (10/12) were: "we are interested in this technology and are studying the possibilities of using it." In one case, the answer was, "we are interested in this technology; we are in the design/implementation stage." Only one of the surveyed cities has implemented digital twins solutions. However, the answer is accompanied by the comment that the area of application is the registration of physical objects in the city, and the scope of implementation is limited and treated rather as an experiment for the moment.

Most cities are at the exploratory or pre-project stage. This means that offices, and especially departments responsible for smart city development, are trying to analyze the potential benefits of the new technology on an ongoing basis. Two-thirds of the responses (8/12) include the answer that the potential of digital twins technology can

already be very useful in supporting management processes. Four forms, however, included the answer that, at this point, however, it is too early to use this potential, and it is more of a future technology.

Also of interest are the responses to the question on potential application areas of digital twins – Fig. 2.

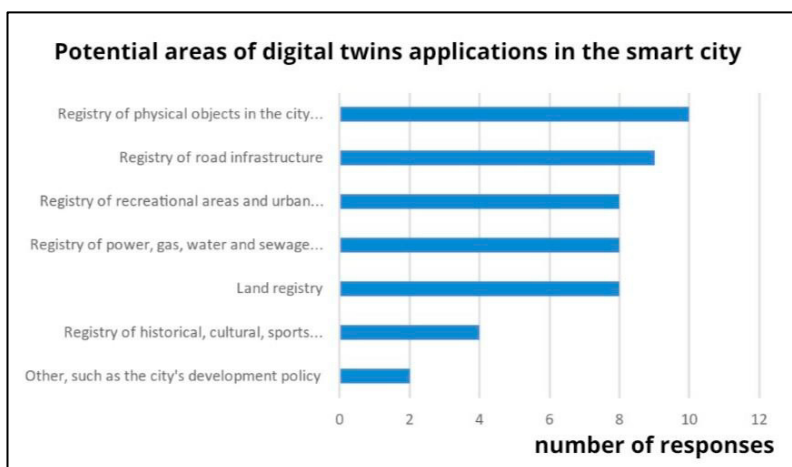


Fig 2. Potential areas of digital twins applications in the smart city.

The chart in Fig. 2 shows that city offices see opportunities to apply digital twins, mainly in recording physical objects in the city. Among the comments added to the forms were opinions that the IT solutions currently in place to support records could be better and have several limitations. The potential of digital twins indicates that these limitations (e.g., in the area of simulation and modeling) can be overcome.

The aspect of simulation and forecasting in the digital environment and the implementation of the results in the physical environment were addressed further in the survey. Respondents indicated that such processes should be implemented mainly in the areas of:

- management of physical facilities in the city (8 responses)
- land management (7 responses)
- management of road infrastructure (9 responses)
- management of energy, gas, water, sewage networks, etc. (7 responses).

Additional comments noted that simulations carried out in a digital environment could prevent costly and often irreversible errors that can occur in urban management. This applies to estimating investment planning of road and communication networks, preventing collisions in power and gas networks, etc. Respondents point out that the solutions available at the moment to support the registration of objects in the city (such as GIS tools, digital land, and building registration systems) do not offer such capabilities.

An essential aspect of the survey of city offices is the expected effects of implementing digital twins technology – Fig. 3.

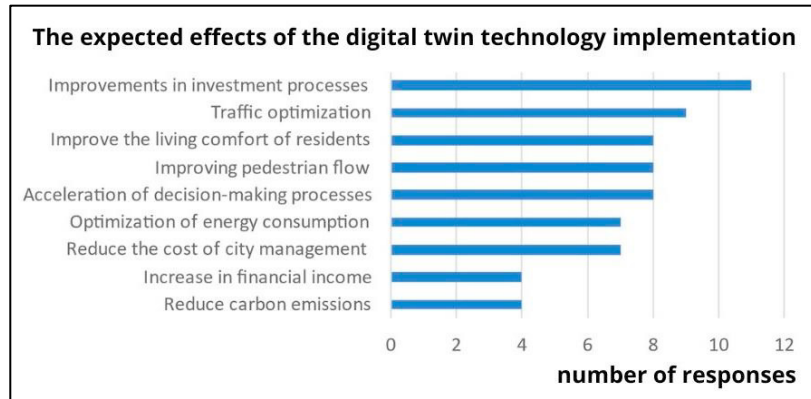


Fig 3. The expected effects of the digital twin technology implementation.

There are some interesting conclusions from the chart in Fig. 3. City halls, despite current trends, place less importance on environmentalism and carbon efficiency than one might expect. Only four respondents indicated effects related to carbon dioxide reduction. Nor do city executives expect to increase city revenues. Respondents' answers mainly relate to improving investment and management processes. This means that the potential of new technologies would attract new investors who will indirectly contribute to the development of cities.

Attention was also paid to the comfort of residents. This comfort depends on the general conditions but is also related to reducing traffic jams and improving overall communication.

While the implementation of digital twin technology in smart cities is expected to bring enhancements across various sectors, such as improved urban planning, traffic optimization, and improved living comfort, it is imperative to carefully consider the accompanying security and ethical issues to ensure these innovations do not compromise residents' privacy, security, and equitable access to city services. To assess the awareness and preparedness of cities' authorities regarding the security and ethical issues associated with urban digital twin technology, the survey encompassed a series of targeted questions to elicit detailed information to highlight specific risks and challenges. Participants from city authorities raised several issues concerning the comprehensive implementation of digital twin technology. Table 1 presents the identified concerns, highlighting the complexity and potential challenges faced in deploying such technologies across urban environments.

Table 1. Risks and challenges related to City Digital Twin implementation

Identified risks and challenges	Characteristics
Exclusion and Inequality	Concerns about the potential for digital twin implementations to exacerbate existing disparities between urban and rural areas, as well as within different segments of the city population. The risk of deepening the digital divide and the need for strategies to ensure equitable access to technology benefits.
Cybersecurity Threats	Susceptibility of digital twin systems to cyberattacks, which could potentially lead to severe disruptions in key city infrastructures and services.
Security Policy Compliance	Establishing comprehensive security policies and managing resource access. difficulties in defining clear policies that adequately protect data and system integrity within an interconnected digital twin framework.
Privacy and Data Leakage	Leakage of sensitive inhabitant data and increased surveillance could lead to public dissatisfaction and trust issues. The balance between leveraging data for urban improvement and safeguarding residents' privacy.
Workforce Impact	Fears of a widening skills gap and job losses in city management and operations. The need for proactive measures, such as training and education programs, to mitigate these effects and facilitate a smooth transition

Dependency on External Providers

Growing dependency on external technology providers could affect the autonomy and resilience of city systems and services. This dependence poses strategic risks, including vendor lock-in and increased operational vulnerabilities.

To explore the depth of awareness and the level of concern among city officials about the critical issues the survey participants were asked to evaluate each concern on the Likert scale where 5 means very serious, and 1 – very low concern. The results of the evaluation are presented in Fig. 4.

City	Cybersecurity	Data Privacy	Dependency on Providers	Workforce Impact	Compliance Issues	Exclusion and Inequality
City 1	5	4	3	2	3	3
City 2	4	5	3	3	5	2
City 3	5	2	4	5	3	4
City 4	4	3	5	4	2	2
City 5	3	5	3	2	4	3
City 6	4	4	2	3	3	4
City 7	5	3	4	4	4	3
City 8	3	2	3	3	2	4
City 9	5	4	5	3	4	2
City 10	4	5	2	4	5	3
City 11	4	3	4	5	2	4
City 12	4	4	3	2	4	2
City 13	5	5	5	4	3	1
City 14	3	5	3	2	1	1
City 15	4	4	3	4	3	2
City 16	5	2	3	4	2	5
TOTAL	67	60	55	54	50	45

Fig 4. Evaluation of concerns by the survey participants.

These concerns reflect a broad spectrum of challenges that city authorities anticipate as they move forward with implementing digital twin technologies. Cybersecurity is the most prominent concern among the cities, with a total score of 67. Most cities rated this issue with high levels of concern (4 or 5), indicating widespread apprehension about potential security vulnerabilities associated with digital twins. Given the geopolitical tensions and the observed increase in cyberattacks linked to Russian actors, there is a heightened risk of targeted cyber espionage or disruptive attacks against critical infrastructure. Digital twins, which integrate extensive data from various city services and infrastructures, could become prime targets of which the respondents are aware.

Data privacy is another major concern, with a total score of 60. This reflects significant worries about handling and protecting sensitive data within digital twin frameworks, closely aligning with cybersecurity issues. The dependency on external technology providers scores a total of 55. While this is a considerable concern, it is slightly less critical than cybersecurity and data privacy, indicating a moderate level of unease about reliance on third-party solutions and services. Digital twins involve complex technologies requiring specialized knowledge and expertise in IoT, AI, data analytics, and cloud computing. Many municipalities lack the in-house capacity to develop these systems from scratch. Therefore, reliance on external providers can indeed seem unavoidable. Moreover, the study participants noticed that maintaining cutting-edge systems can be challenging for city governments. External providers dedicate resources specifically to innovation and staying current with technology trends, which can be a significant advantage.

Concerns about the impact of digital twins on the workforce total 54. This reflects worries about job displacement and the need for new skills, suggesting that cities are aware of the social implications of technological implementation. As digital twins automate processes that were previously done manually, there's a risk of job displacement for workers who perform those tasks. This can include roles in monitoring, maintenance, and management of city infrastructure, where digital twins can optimize operations to require fewer human inputs, which can change traditional work environments.

Compliance with legal and regulatory frameworks scores 50, the second lowest of the concerns. This suggests a moderate awareness of the challenges posed by regulations like GDPR, but it is not seen as the most pressing issue. Given existing laws around data protection, privacy, and IT security, many cities already have some infrastructure for dealing with compliance. The expertise and processes established for these purposes can often be extended or adapted to new technologies like digital twins.

With the lowest total score of 45, concerns about exclusion and inequality indicate some awareness of the potential social disparities digital twins might exacerbate. However, it appears to be the least urgent concern among the surveyed issues. Since digital twin projects are typically urban-centric, concerns specific to rural or underserved urban areas might be deprioritized or overlooked in the planning stages. Moreover there might be an assumption that advancements in IT infrastructure inherently benefit all segments of society equally, leading to a perception that technology-driven solutions will naturally close rather than widen gaps.

5. Conclusions

As the study shows, implementing smart city digital twins in Poland is definitely possible. The authorities of the largest cities know these complex systems' benefits and potential. The implementation of digital twins requires the use of many technologies that are already well-known and used in Poland. There are many software and infrastructure providers. Many companies specialize in Internet of Things (IoT) devices, data analytics, and cloud computing. This can be achieved through partnerships with technology providers and investment in necessary infrastructure.

Creating a smart city digital twin requires the utilization of existing data sources. This can include data from sensors, cameras, and other urban infrastructure, as well as legacy systems operating in city offices. By aggregating and analyzing this data, a detailed virtual model of the city can be created. The challenge of heterogeneous data integration seems crucial for achieving the analytical goals of the smart city digital twin.

The possibilities of semantic linkage between IoT networks, management information systems, and web resources for supporting intelligent city management are the next step in the evolution of smart city digital twins. Since the implementation of digital twins is a multi-stage project and the execution is incremental, it is important that cities have clearly defined social, ecological, and economic goals. Cost-benefit analysis and social consultations should precede the selection of priorities for project implementation. Engagement with local stakeholders is crucial for the success of any smart city initiative. This includes city officials, residents, and businesses. By involving these stakeholders in the planning and implementation process, the digital twin can be customized to meet the needs and priorities of the community.

Furthermore, as Poland experiences a notable increase in cyber threats, particularly from actors sponsored by foreign states, the need for robust cybersecurity measures cannot be overstated. Implementing advanced security protocols and ensuring compliance with national and international data protection regulations will be essential to safeguard these digital assets. Similarly, addressing the ethical considerations of surveillance, data privacy, and potential workforce displacement will be critical to maintain public trust and ethical integrity of the implementation process.

In the presented study, the authors focused on the potential of using digital twins in Polish cities. A very positive phenomenon is the functioning of organizational units responsible for recognizing modern IT solutions. The further intention of the authors is to observe the development of the implementation of digital twin technology in Poland and the world. Soon it will be possible to critically assess the activities of local governments in this area and identify best practices. The findings from this survey are expected to contribute to the broader discourse on the safe and responsible deployment of digital twin technologies in urban environments, providing a benchmark for other cities and guiding future policy development.

References

- [1] Kim, S.C. , P. Hong, T. Lee, A. Lee, and S. H. Park. (2022) "Determining Strategic Priorities for Smart City Development: Case Studies of South Korean and International Smart Cities." *Sustainability* **14**, 10001. <https://doi.org/10.3390/su141610001>.

- [2] Brzeziński, Ł. and M.K. Wyrwicka. (2022) “Fundamental Directions of the Development of the Smart Cities Concept and Solutions in Poland.” *Energies* **15**(21) 8213. <https://doi.org/10.3390/en15218213>.
- [3] Jelonek, D., A. Mesjasz-Lech, C. Stepniak, T. Turek and L. Ziora. (2020) “Potential data sources for sentiment analysis tools for municipal management based on empirical research.” *Lecture Notes in Networks and Systems*, **69**, pp. 708 – 724.
- [4] Turek, T. and C. Stepniak. (2021) “Areas of integration of GIS technology and smart city tools. Research findings.” *Procedia Computer Science*, **192**, pp. 4681 – 4690.
- [5] Turek, T. and C. Stepniak. (2022) “Barriers to the Application of Spatial Information Systems in the SmartCity Dynamic Management.” *Procedia Computer Science*, **207**, pp. 4217 - 4226
- [6] Deren, L., Y. Wenbo and S. Zhenfeng. (2021) “Smart city based on digital twins.” *Computational Urban Science*. **1**, 4. <https://doi.org/10.1007/s43762-021-00005-y>
- [7] Enders, M. R. and N. Hoßbach. (2019) “Dimensions of digital twin applications-a literature review.” *25th Americas Conference on Information Systems*, AMCIS 2019.
- [8] Lei, B., Stoter, J. E., Janssen, P., & Biljecki, F. (2023). Uncovering the challenges of urban digital twins: Identifying and ranking the barriers to operating digital twins in cities. GIM International: the worldwide magazine for geomatics, 2023(June). <https://www.gim-international.com/content/article/uncovering-thechallenges-of-urban-digital-twins>
- [9] Botín-Sanabria, D.M., Mihaita, A., Peimbert-García, R.E., Ramírez-Moreno, M.A., Ramírez-Mendoza, R.A., & Lozoya-Santos, J.D. (2022). “Digital Twin Technology Challenges and Applications: A Comprehensive Review.” *Remote Sensing*, **14**, 1335. <https://doi.org/10.3390/rs14061335>
- [10] Nalini, D. (2024) “Securing Smart Cities: A Cybersecurity Perspective on Integrating IoT, AI, and Machine Learning for Digital Twin Creation.” *Journal of Electrical Systems*. **50**, 5s <https://doi.org/10.52783/jes.3052>.
- [11] OECD (2019) “Enhancing Access to and Sharing of Data: Reconciling Risks and Benefits for Data Re-use across Societies.” *OECD Publishing*, Paris, <https://doi.org/10.1787/276aaca8-en>.
- [12] He, L., G. Wu, D. Dai, L. Chen, and G. Chen. (2011) “Data Conversion between CAD and GIS in Land Planning.” *In Proceedings of the 2011 19th International Conference on Geoinformatics*, Shanghai, China, 24–26 June 2011; pp. 1–4.
- [13] Qian, C., X. Liu, C. Ripley, M. Qian, F. Liang and W. Yu (2022). “Digital Twin - Cyber Replica of Physical Things: Architecture, Applications and Future Research Directions.” *Future Internet*, **14**, 64. <https://doi.org/10.3390/fi14020064>
- [14] Dimitrov, H. and D. Petrova-Antonova, 3D city model as a first step towards digital twin of Sofia City. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.-ISPRS Arch.*, **43**, 23–30. 2021 <https://doi.org/10.5194/isprs-archives-XLIII-B4-2021-23-2021>
- [15] Ladj, A., Z. Wang, O. Meski, O., F. Belkadi, M. Ritou and C. Da Cunha. (2021) “A knowledgebased digital shadow for machining industry in a digital twin perspective.” *J. Manuf. Syst.* **58**, 168–179. <https://doi.org/10.1016/j.jmsy.2020.07.018>
- [16] Gassmann, O., J. Bohm, and A. Palmie. (2019) “Smart Cities: Introducing Digital Innovation to Cities.” *Emerald Publishing* p.125 -126
- [17] Boyes, H. and T. Watson. (2022) “Digital Twins: An Analysis Framework and Open Issues.” *Computers in Industry*, **143**, 103763, 2022. <https://doi.org/10.1016/j.compind.2022.103763>
- [18] Schrotter, G. and C. Hürzeler. (2020) “The Digital Twin of the City of Zurich for Urban Planning.” *Journal of Photogrammetry, Remote Sensing and Geoinformation Science*, **88**, pp.99–112. <https://doi.org/10.1007/s41064-020-00092-2>
- [19] Boje, C., A. Guerriero, S. Kubicki and Y. Rezgui. (2020) “Towards a semantic Construction Digital Twin: Directions for future research.” *Automation in Construction*, **114**, 103179, 2020. <https://doi.org/10.1016/j.autcon.2020.103179>
- [20] Ravid, B. Y. and M. Aharon-Gutman. (2022) “The Social Digital Twin: The Social Turn in the Field of Smart Cities.” *Environment and Planning B: Urban Analytics and City Science*, **50** (6). <https://doi.org/10.1177/23998083221137079>
- [21] White, G., A. Zink, L. Codecá and S. Clarke. (2021) “A digital twin smart city for citizen feedback.” *Cities* **110**, 103064, <https://doi.org/10.1016/j.cities.2020.103064/>
- [22] Helbing, D. and J. A. Sánchez-Vaquerizo. (2022) “Digital Twins: Potentials, Ethical Issues, and Limitations.” *In E. Elgar (eds.): Handbook on the Politics and Governance of Big Data and Artificial Intelligence*, Forthcoming. <http://dx.doi.org/10.2139/ssrn.4167963>
- [23] Wan, L., T. Nohta and J.M. Schooling. (2019) “Developing a City-Level Digital Twin - Propositions and a Case Study.” *International Conference on Smart Infrastructure and Construction 2019 (ICSIC)*. January 2019, pp. 187-194 <https://doi.org/10.1680/icsic.64669.187>
- [24] Dembski, F., U. Wössner, M. Letzgus, M. Ruddat, and C. Yamu. (2020) “Urban Digital Twins for Smart Cities and Citizens: The Case Study of Herrenberg, Germany.” *Sustainability*, **12**, 2307. <https://doi.org/10.3390/su12062307>